

Ken Nakamura

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RESEARCH FOCUS

I study how artificial neural networks and biological brains acquire and organize internal representations through learning. Building on these insights, I seek to identify what makes representations efficient and robust for specific problems. I also aim to understand the brain's dynamic information-processing mechanisms and develop computational architectures inspired by these principles.

EDUCATION

The University of Tokyo, Faculty of Engineering

Apr 2023–Present

B.Eng., Mathematical Engineering and Information Physics

Expected graduation: March 2027 (03/27). Granted an exemption from the 30-credit-per-semester cap for academic excellence; one of approximately 10 recipients in my entering class.

PUBLICATIONS AND PREPRINTS

- [1] **Nakamura, K.**, Nakai, T., Yashiro, R., Yamashita, A., & Amano, K. (2026). Beyond Prediction Accuracy: Target-Space Recovery Profiles for Evaluating Model-Brain Alignment. *arXiv preprint*, arXiv:2605.20127. Under review.
- [2] Asanuma, H., Koide-Majima, N., **Nakamura, K.**, Horii, T., Nishimoto, S., & Oizumi, M. (2025). Correspondence of high dimensional emotion structures elicited from video clips between humans and multimodal LLMs. *Scientific Reports*, 15, 32175. doi:10.1038/s41598-025-14961-6.

RESEARCH EXPERIENCE

Amano-Nakai-Nakayama Lab, The University of Tokyo

Jun 2024–Present

Undergraduate Researcher

Tokyo, Japan

Advisors: Prof. Kaoru Amano and Assoc. Prof. Tomoya Nakai

- Proposed target-space recovery profiles to identify which reproducible dimensions of human fMRI responses are captured by vision models and other human brains.
- Showed that similar prediction accuracy can mask differences in recovered brain-response structure.

German Research Center for Artificial Intelligence (DFKI)

Feb 2026–Apr 2026

Guest Researcher, Smart Data & Knowledge Services Department

Kaiserslautern, Germany

Advisor: Prof. Andreas Dengel

- Studied LLM self-distillation in tool-use and mathematical-reasoning settings.
- Built training and evaluation protocols to analyze learning instability.

Sulchek BioMEMS and Biomechanics Lab, Georgia Tech

Aug 2025–Sep 2025

Visiting Research Fellow

Atlanta, GA, USA

Advisor: Prof. Todd Sulchek

- Curated over 23,000 cell-morphology images from high-speed microfluidic compression videos, including segmentation masks and normal/abnormal morphology labels.
- Trained models and built an image-analysis pipeline to detect abnormal cell morphology.

Oizumi Lab, The University of Tokyo

Apr 2024–Aug 2025

Undergraduate Researcher

Tokyo, Japan

Advisor: Prof. Masafumi Oizumi

- Collected emotion ratings from multimodal LLMs for naturalistic videos.
- Compared model and human emotion representations using representational similarity analysis and Gromov–Wasserstein optimal transport.

RESEARCH EXPERIENCE (CONTINUED)

Ikeuchi Lab, The University of Tokyo

Undergraduate Research Assistant, Institute of Industrial Science

Advisor: Prof. Yoshiho Ikeuchi

Oct 2023–Jan 2024

Tokyo, Japan

- Contributed to stimulation design and data collection for experiments using electrical feedback to train in vitro mouse cerebral networks on maze-navigation tasks.

SELECTED PROJECTS

Multilingual and Mathematical Typesetting Engine

MITOU IT Program | Rust, Tauri

- Built a typesetting engine from scratch in Rust for multilingual text and mathematical notation.
- Implemented real-time preview and GUI-based editing directly in the preview using Tauri.

RESEARCH PRESENTATIONS

- [1] **Nakamura, K.**, & Frolov, S. (Apr 2026). *Toward Efficient Continual Learning for LLMs: Two Burdens of Self-Distillation*. Research webinar, DFKI, Kaiserslautern, Germany.
- [2] **Nakamura, K.**, Amiri, H. A., Chen, S., & Sulchek, T. (Sep 2025). *BlebNet: a Cell Damage Detection Network for High-speed Microfluidic Compressions*. Poster, Nakatani RIES Symposium, Georgia Institute of Technology, Atlanta, GA, USA.
- [3] **Nakamura, K.**, Asanuma, H., & Oizumi, M. (Sep 2024). *Emotion Prediction During Video Viewing Using Multimodal LLMs*. Seminar presentation, The University of Tokyo, Tokyo, Japan.

FELLOWSHIPS AND GRANTS

MITOU IT Program, Selected Developer

2025–2026

Ministry of Economy, Trade and Industry (METI), Japan. One of 21 projects selected from 227 applications; JPY 2.88 million in project funding.

Nakatani Advanced Research Internship Program

Oct 2025

Nakatani Foundation. One of 5 Japanese students selected for a fully funded research stay at DFKI.

Nakatani Research and International Experiences for Students Fellowship Program

Mar 2025

Nakatani Foundation / Georgia Tech. One of 11 Japanese fellows selected from 130 applicants for a fully funded research stay at Georgia Tech.

KEYENCE Foundation Student Support Grant

Jun 2026

KEYENCE Foundation.

ASJ Foundation Scholarship

Jun 2025

ASJ Foundation. One of 12 undergraduate recipients in 2025.

HONORS AND AWARDS

MITOU Super Creator Certification, METI

Jun 2026

One of 23 Super Creators certified among 34 creators completing the 2025 MITOU IT Program.

Chemistry Grand Prix Awards, Chemical Society of Japan

2021–2022

Bronze Prize (Aug 2022; approximately top 2.5% of 3,215 participants); Kanto Branch Chief's Award (Nov 2021).

Valedictorian, Seiko Gakuin Junior and Senior High School

Mar 2023

TECHNICAL SKILLS

Programming	Python, Rust, C/C++, MATLAB, JavaScript, TypeScript
Machine Learning	PyTorch, TensorFlow
Application Development	Tauri, React, Next.js
Development Tools	Docker, Git

LANGUAGES

Japanese (Native); English (Fluent; TOEFL iBT 99/120).